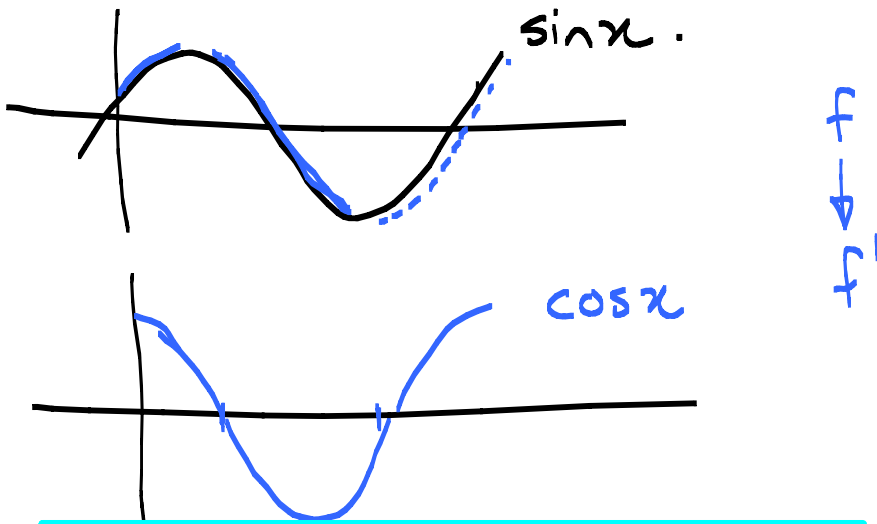


Unit 3.3 Derivatives of Trigonometric Functions.

Note Title

2016/04/12

① $\frac{d}{dx} \sin x = \cos x$ [Know proof p 192]



② $\frac{d}{dx} \cos x = -\sin x$ [Know proof].

$$\frac{d}{dx} \cos x = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{h \rightarrow 0} \frac{\cos x \cosh - \sin x \sinh - \cos x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos x (\cosh - 1)}{h} - \frac{\sin x \sinh}{h}$$

$$= \cos x \left[\lim_{h \rightarrow 0} \frac{\cosh - 1}{h} \right] - \sin x \left[\lim_{h \rightarrow 0} \frac{\sinh}{h} \right]$$

$= 0$ $= 1$

$$= -\sin x \rightarrow$$

$$\begin{aligned}
 * \lim_{h \rightarrow 0} \frac{\cosh h - 1}{h} & \quad \left(\frac{0}{0} \right) \quad \left(\frac{\cosh h + 1}{\cosh h + 1} \right) \\
 &= \lim_{h \rightarrow 0} \frac{\cosh^2 h - 1}{h(\cosh h + 1)} = \lim_{h \rightarrow 0} \frac{-\sinh^2 h}{h(\cosh h + 1)} \quad \left(\frac{0}{0} \right) \\
 &= \lim_{h \rightarrow 0} \frac{\sinh h}{h} \cdot \frac{-\sinh h}{\cosh h + 1} \\
 &= 1 \cdot \frac{0}{1+1} = 0
 \end{aligned}$$

While we are waiting:

$$\textcircled{1} \frac{d}{dx} \frac{\sin x}{(1 + \cos x)} = \frac{\cos x(1 + \cos x) - \sin x(-\sin x)}{(1 + \cos x)^2}$$

$$\begin{aligned}
 \textcircled{2} \frac{d}{dx} e^x \cdot \cos x &= \frac{\cancel{\cos x} + 1}{(1 + \cos x)^2} = \frac{1}{1 + \cos x} \\
 &= e^x \cos x + e^x(-\sin x)
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{3} \frac{d}{dx} \operatorname{cosec} x &= \frac{d}{dx} \frac{1}{\sin x} \\
 &= \frac{0 - \cos x}{(\sin x)^2}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{4} \frac{d}{dx} \tan x &= -\cot x \cdot \operatorname{cosec} x \\
 &= -\operatorname{cosec} x \cdot \cot x
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{4} \quad \frac{d}{dx} \tan x &= \frac{d}{dx} \frac{\sin x}{\cos x} \\
 &= \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} \\
 &= \frac{1}{\cos^2 x} = \sec^2 x
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{5} \quad \frac{d}{dx} \sec x &= \sec x \tan x \\
 \textcircled{6} \quad \frac{d}{dx} \cot x &= -\operatorname{cosec}^2 x
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{7} \quad y &= \pi \sec x + \cot x \\
 y' &= \pi \sec x \tan x + (-\operatorname{cosec}^2 x) !!!
 \end{aligned}$$

$$\textcircled{8} \quad y = \frac{\operatorname{cosec} x}{e^x}, \quad y' = \frac{-\operatorname{cosec} x \cot x (e^x) - \operatorname{cosec} x (e^x)}{(e^x)^2}$$

$$\begin{aligned}
 \textcircled{9} \quad \frac{d}{dt} [\sqrt{t} + \tan t] \\
 = \frac{1}{2\sqrt{t}} + \sec^2 t.
 \end{aligned}$$